Session plan 5: Descriptive inference

Thu 08.10.26, 16:15–17:45 · Room 4.B51 · Basic reading: KW ch. 6 (pp. 125–141); additional: KKV ch. 2, Gerring (2012), de Kadt & Grzymala-Busse (2025) · Applied reading: Barnes & Karim (2025), *The Manosphere and politics* · Presenters: **Rika Elena Hünerbach, Samantha Irving**

Learning goals: students can (1) classify variables as categorical, ordinal or continuous; (2) choose central tendency and spread measures by variable type and compute them in R; (3) read a histogram and a box plot; (4) name the criteria for good estimators; (5) derive descriptive questions from a theory.

Before class: presentation slides in OLAT (due Wed 07.10, 12:00)? · seminar question 5 uploaded? · grading sheet printed · deck rendered with the 2026 class list (`00_admin/classroom_2026.csv`) · deck in speaker view (press S)

Checkpoints: 16:35 start "Central tendency" · 16:55 start "Good descriptive inference" · **17:07 end share round** · **17:08 presentation starts** (25 min max., then discussion until 17:42) · **Content note** before the presentation: misogyny and violence against women

Time	Slide	Content	Interaction	Guiding questions	If behind
16:15–16:17	2	Four components of a research design (KKV p. 13); today: the use of the data. Sessions 5–7: descriptive, causal, predictive	Input		
16:17–16:21	3	Descriptive inference = understanding the unobserved from observations (KKV p. 55); description is not "mere" (KKV p. 34)	Input + question	What is the step from describing to *inferring*?	
16:21–16:22	4	One variable vs. two or more	Input		
16:22–16:24	5	Our classroom: population (UniLu students), sample (this class), gender and name length	Question to the room	Can we generalise from this room to all UniLu students?	
16:24–16:29	6	**Variable types**: categorical, ordinal, continuous (KW pp. 126–130); reveal gender = categorical, name length = continuous	Students give examples	Give an example of each type. Which ordinal variables do you know?	
16:29–16:30	7	Data listing (tibble)	Input	What is a typical name length?	
16:30–16:32	8	Bar plot for categorical variables; bars over pies (KW p. 131)	Input	Why a bar plot and not a pie chart?	
16:32–16:35	9	Histogram (bin width 2) and density curve; bins change the impression (KW Fig. 6.6)	Input; click between the bin width 2 and 5 tabs	Does the shape change with the bin width? Which would you report?	Skip the bin width 5 tab
16:35–16:39	10	**Central tendency** table: mode, median, mean by variable type	Ask first, then reveal	Which measures do you know? What does each one need from the variable?	
16:39–16:42	11–12	Mode of gender; mean and median of name length (code + result)	Demo	Why not the mean of gender? What if one name had 60 letters?	Results only
16:42–16:45	13	**Spread** table: range, Q1/Q3, IQR, variance, SD by variable type	Ask first, then reveal	Which measures of spread do you know? Which need equal distances?	
16:45–16:47	14	Spread for name length (quartiles with `type = 2` = KW's method: Q1 = 12.5)	Demo	Why square the deviations?	Results only

Time	Slide	Content	Interaction	Guiding questions	If behind
16:47–16:49	15	Box plot = the numbers just computed; 1.5 × IQR outlier rule (KW pp. 135–136)	Students match parts to numbers	Where are the median, Q1 and the IQR?	
16:49–16:51	16	Summary: centre, spread, shape; frequencies, ranks, moments	Input		30 seconds
16:51–16:53	17	**Your turn**: visualise gender × name length (hint: categorical × continuous)	Think, then 2–3 answers	How would you plot it? Why is a bar chart of means weaker?	
16:53–16:55	18	Box plots by gender: an association, not a cause	Input + question	Real difference or noise?	
16:55–16:58	19	Good estimators: unbiased, efficient, consistent (KKV pp. 63–67), linked to session 4	Input + question	Which session 4 idea matches each criterion?	
16:58–17:00	20	From description to research questions: five questions from economic voting (session 3)	Input	Which of these questions are descriptive?	
17:00–17:02	21	Ladder of research questions (de Kadt & Grzymala-Busse, Fig. 1)	Map the questions onto the rungs	On which rung is your own question?	
17:02–17:07	22	**Share your descriptive question** (seminar question 5)	2–3 volunteers	Which rung? Does it reflect, build or evaluate the theory? Is it really descriptive?	Fewer volunteers; **hard stop 17:07**
17:07–17:08	23	Hand over; content note			
17:08–17:33	—	**Student presentation** (25 min max.): summary and critique ~10 min, alternative design ~15 min	Presentation; grade with the sheet		Signal at 20 min
17:33–17:42	—	**Discussion** to improve the proposal (prompts below)	Plenary	See prompts below	
17:42–17:45	24–26	Session 6 on 15.10, 14:15, room 3.B48 (BF ch. 2–3; Lindegger & Schär); seminar questions 6–7 due Wed 14.10, 12:00; take-home exercise I published 15.10	Admin		

Discussion prompts (if the room is quiet): Snowball sampling from one starting list (SPLC): which population can the 140 sites describe? · What is left out (4chan, private channels, non-English sites)? · Intercoder agreement of 75.9% on subgroups: reliable enough (session 4)? · The typology is built from the same data it is meant to support: what would an independent test look like? · Which rung of the ladder does the paper reach?

Grading (0–5 each, 15/30 = pass): clarity · understanding · critique quality · reasoning · feasibility · discussion. Both presenters, one grade.

Notes during class: